

Team Worksheet — Lab 24

Fractional Factorial Designs, Confounding and Resolution (Black Belt Only)

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-24-fractional-factorial-doe.xlsx`.

3. What we produced

Deliverable: A fractional factorial screening design with a documented confounding structure.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Understand the economics: 2^7 full factorial = 128 runs; a $2^{(7-4)}$ fractional = 16 runs. The saving is real but it is paid for in resolving power.

Step 2. Understand confounding (aliasing): in a fractional design, some effects are mathematically indistinguishable from others.

Step 3. Learn the generator and defining relation — the algebra that determines exactly which effects are aliased with which.

Step 4. Learn the resolution scale precisely. RESOLUTION III: main effects aliased with two-factor interactions. RESOLUTION IV: main effects clear of two-factor interactions, but two-factor interactions aliased with each other. RESOLUTION V: main effects and two-factor interactions all clear.

Step 5. Understand the selection rule: use Resolution III only for early screening of many factors; use IV or V when you intend to interpret interactions.

Step 6. Select 5 or more candidate factors from your capstone for a screening experiment.

Step 7. Choose your fraction based on your run budget, and write down the resulting resolution BEFORE running anything.

Step 8. Generate the design matrix and write out the complete alias structure. You must be able to state what each estimated effect is confounded with.

Step 9. Apply effect sparsity and hierarchy: usually few factors dominate, and low-order effects are more likely to be real than high-order ones.

Step 10. Randomise and execute the screening experiment.

Step 11. Analyse the effects and identify the vital few factors. Where an important effect is confounded, state the ambiguity explicitly rather than assuming the main effect is the real one.

Step 12. Resolve critical ambiguity with a fold-over design — a second fraction that de-aliases the effects you care about.

Step 13. Take the screened vital few factors forward to a full factorial or RSM study in Lab 25.

7. Self-check

You stated the design resolution before running, wrote out the complete alias structure, and any ambiguous conclusion is either declared explicitly or resolved by a fold-over.

Complete? ■ Yes ■ Not yet

